

Week 4

Vibe Coding, Interaction Design, and Academic Research Support

Seminar Agenda



Part 1: Technical Guidance & Interaction Design Learn the basics of "vibe coding" and how to effectively prompt AI tools to design specific, meaningful interaction points.



Part 2: Research & Academic Support Discover how to justify your design decisions using scholarly articles and learn to extract value from research papers.



Part 3: Prototype Deployment & Demos Understand how to launch your AI-generated code to platforms and properly configure a local testing environment.

Part 1: Technical Guidance

Bridging the gap between conceptual designs and interactive prototypes using AI.

Vibe Coding for Beginners

- **Natural Language:** Describe your desired UI/UX in plain English. The AI handles the underlying boilerplate syntax.
- **Iterative Prompting:** Start with a broad structure (the "vibe"), then chat with the AI to refine specific interactions step-by-step.
- **Context is King:** Provide the AI with clear design systems, user personas, and visual references to get better results.
- **Tools:** Utilize AI-first IDEs like Cursor or Windsurf, or web platforms like v0 and Bolt.new for instant visual feedback.



Prompting for Interaction



Micro-Interactions

Prompt the AI to include subtle visual feedback. Instruct it: "Add smooth hover states to buttons and a subtle loading spinner during transitions."



Clear Affordances

Specify visual cues in your prompts. Tell the AI: "Ensure all clickable elements look elevated and interactive, matching standard UI patterns."



Error Prevention

Guide the AI to build safeguards. Request: "Implement real-time form validation and disable the submit button until all required fields are correct."

Part 2: Research & Support

Grounding your AI-assisted designs in scientific evidence and HCI literature.

Academic Requirement

5

Research Papers

Justifying Your Decisions

Even though AI helps build the prototype rapidly, your design decisions cannot be arbitrary. Each student must submit and reference five peer-reviewed articles.

These papers serve as evidence to prove your prompt directions and interaction choices are grounded in proven HCI methodologies.

Finding Relevant Research

- Use academic databases like **Google Scholar**, **ACM Digital Library**, and **IEEE Xplore** for verified literature.
- Search with specific, targeted keywords: "usability," "interaction design," "cognitive load," or specific UI patterns.
- Focus on recent publications (last 5 years) to ensure your evidence reflects modern computing standards and user behaviors.



Analyzing a Research Paper

1. Abstract

Read this first to quickly determine if the paper is highly relevant to the UI concepts you are generating.

2. Introduction

Understand the problem space, the researchers' motivation, and the gap in current design literature.

3. Methodology

Identify how they tested their interaction hypothesis.
Look at their user testing methods and tasks.

4. Discussion

Extract the key takeaways.
Turn these insights into concrete prompts for your vibe-coding sessions.

Part 3: Deployment & Demos

Taking your AI-generated code and sharing it effectively with the world.

Deployment Solutions



Integrated Hosting

Platforms like v0, Lovable, or Bolt.new allow you to deploy your generated UI directly from the chat interface with one click.



Vercel

Offers a lightning-fast global edge network. Seamlessly integrates with AI tools for modern web frameworks like Next.js.



Netlify

Provides automated builds. If you exported your AI project locally, simply drag and drop your project folder to go live.

Conducting a Local Demo

- **Environment Setup:** Install an AI-powered code editor like Cursor or Windsurf.
- **Local Server:** Ask the built-in AI agent (e.g., Cursor Composer) to start your development server in the terminal.
- **Troubleshooting:** If you hit an error before a demo, paste the error into the AI chat to auto-fix broken paths or dependencies.
- **Responsiveness:** Prompt the AI to ensure the design is mobile-friendly, then test across different browser window sizes.



Questions?

Thank you for attending the seminar.